GlucoTwin India

An adaptive patient twin for glucose forecasting

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Digital Twin Challenge 2026 · Research proof-of-concept

A focused clinical question

Forecast a measurable outcome

Glucose at 30, 60 and 120 minutes
Primary benchmark: 60-minute mean absolute error

Exploratory adverse-event endpoint

New onset above 180 mg/dL for three five-minute bins, all inside the next hour.

Intended user

A clinician reviewing a patient trajectory, uncertainty and data quality.

A forecast horizon is not the same as achieved warning lead time.

Two streams, one participant

45
study participants

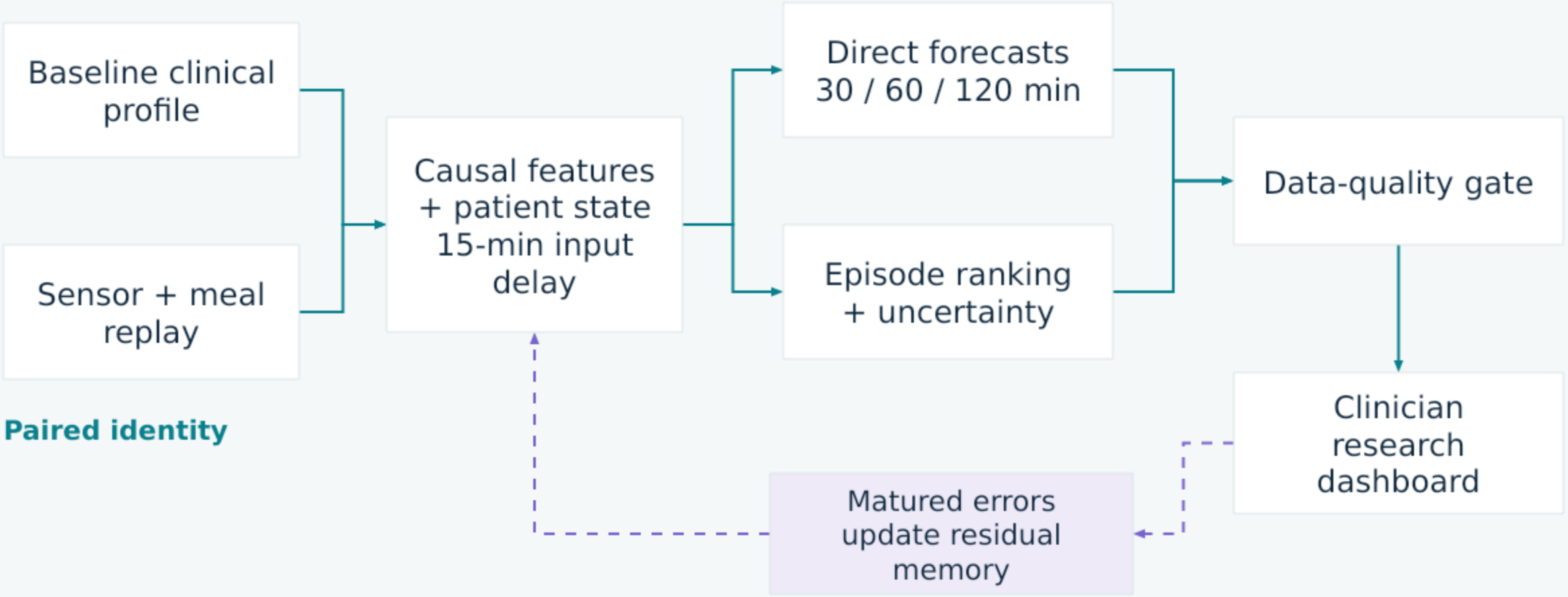
14
in the T2D cohort

US cohort
Indian validation still required

Stream	Fields used	Linkage
Clinical profile	Age, sex, BMI, HbA1c, fasting glucose	Same participant ID
Dynamic observations	Glucose, heart rate, activity METs	Same participant ID
Recorded context	Meal carbohydrate, protein and fat	Same participant ID

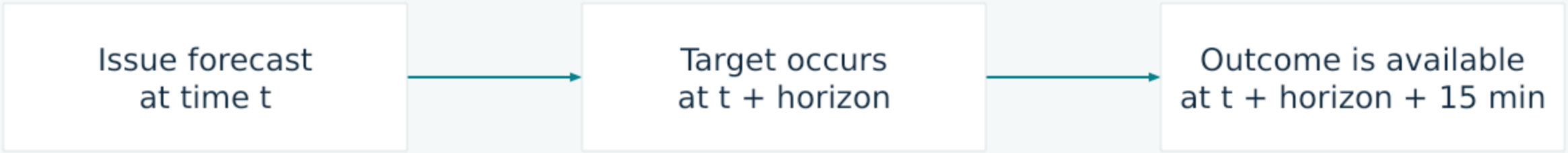
CGMacros v1.0.0 · CC BY-NC-SA 4.0 · EHR-like records; no hospital integration

Architecture and update loop



Scenario sandbox is separate from evaluated forecasts.

Adapt only from the observable past



No future feedback

A 60-minute forecast can update memory only after 75 minutes.

Prefix-invariance tests

Changing the future must not change earlier features or forecasts.

Source limitation remains

Upstream interpolation cannot be fully undone without native device exports.

Evaluation before presentation

Partition	Participants	Purpose
Training	36	Fit population models
Validation	3 T2D	Select variants and alert threshold
Calibration	3 T2D	Estimate residual interval widths
Final test	3 T2D	Report held-out performance

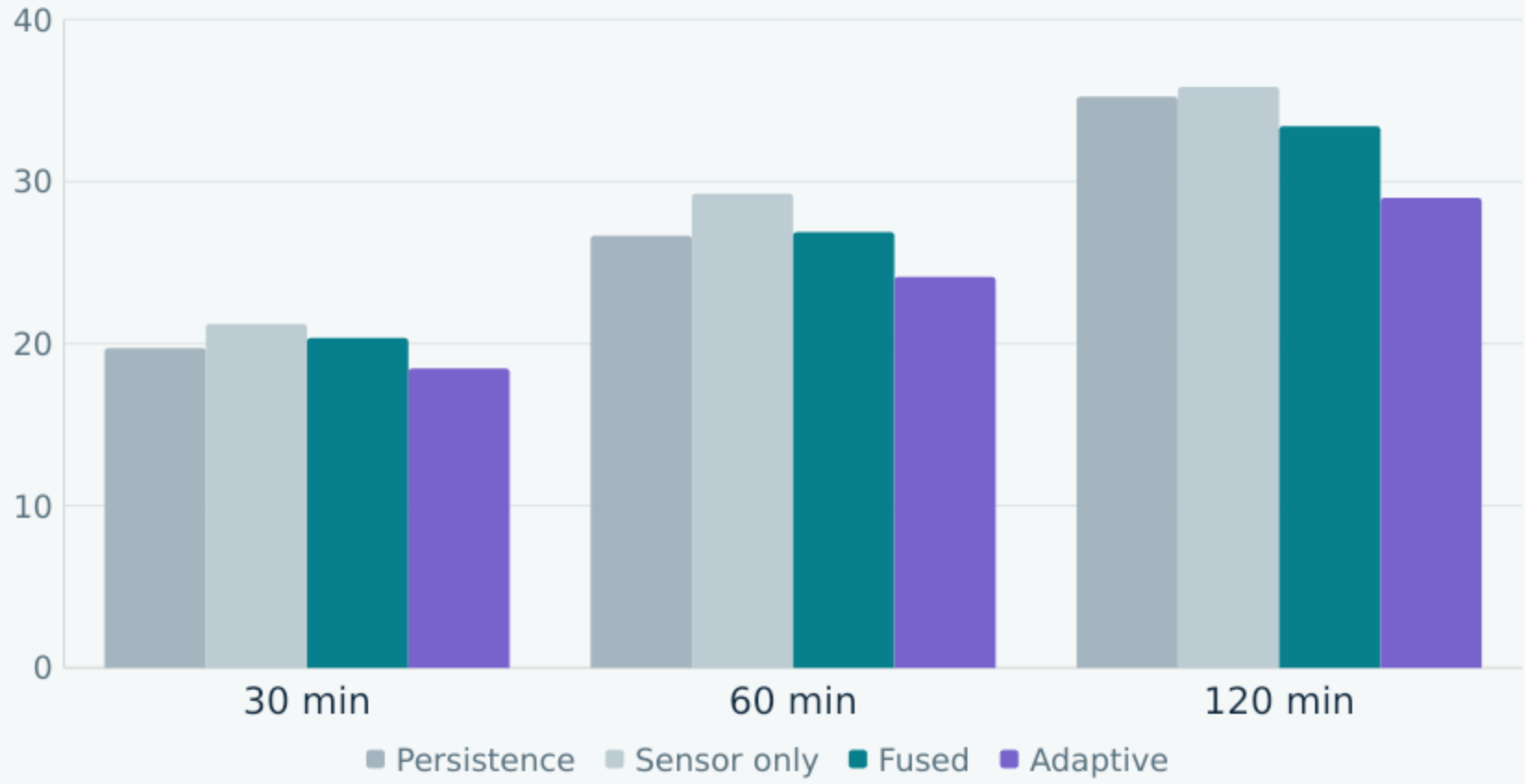
24-hour warm-up

Earlier observations may personalize the twin; future observations cannot.

Patient-macro metrics

Each test participant contributes equally to headline forecast errors.

Measured forecast performance



9.6%

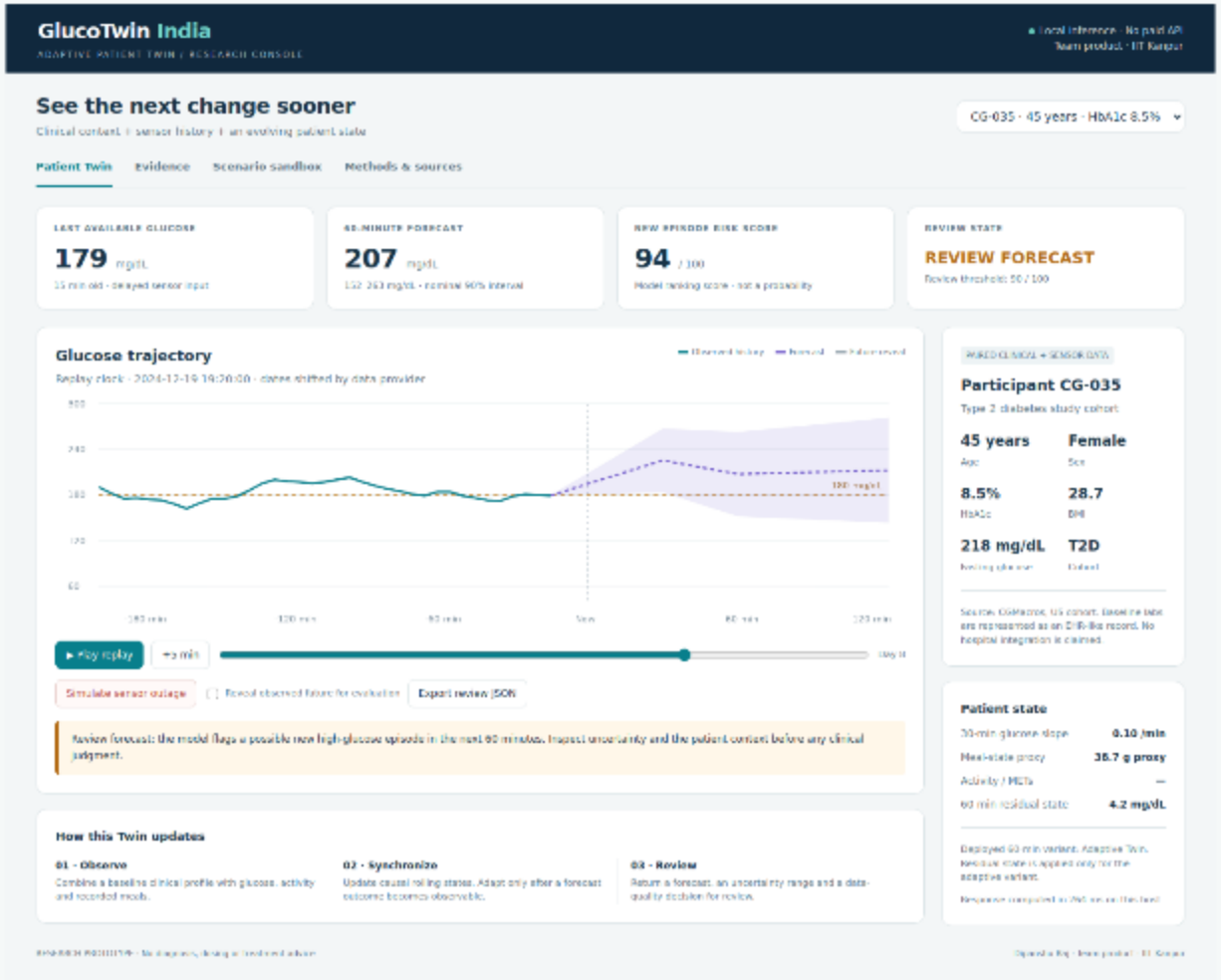
lower MAE at 60 min

17.7%

lower MAE at 120 min

MAE, mg/dL · 3 held-out T2D participants · Fused alone does not win at every horizon

Working prototype walkthrough



Replay

Step through observations.

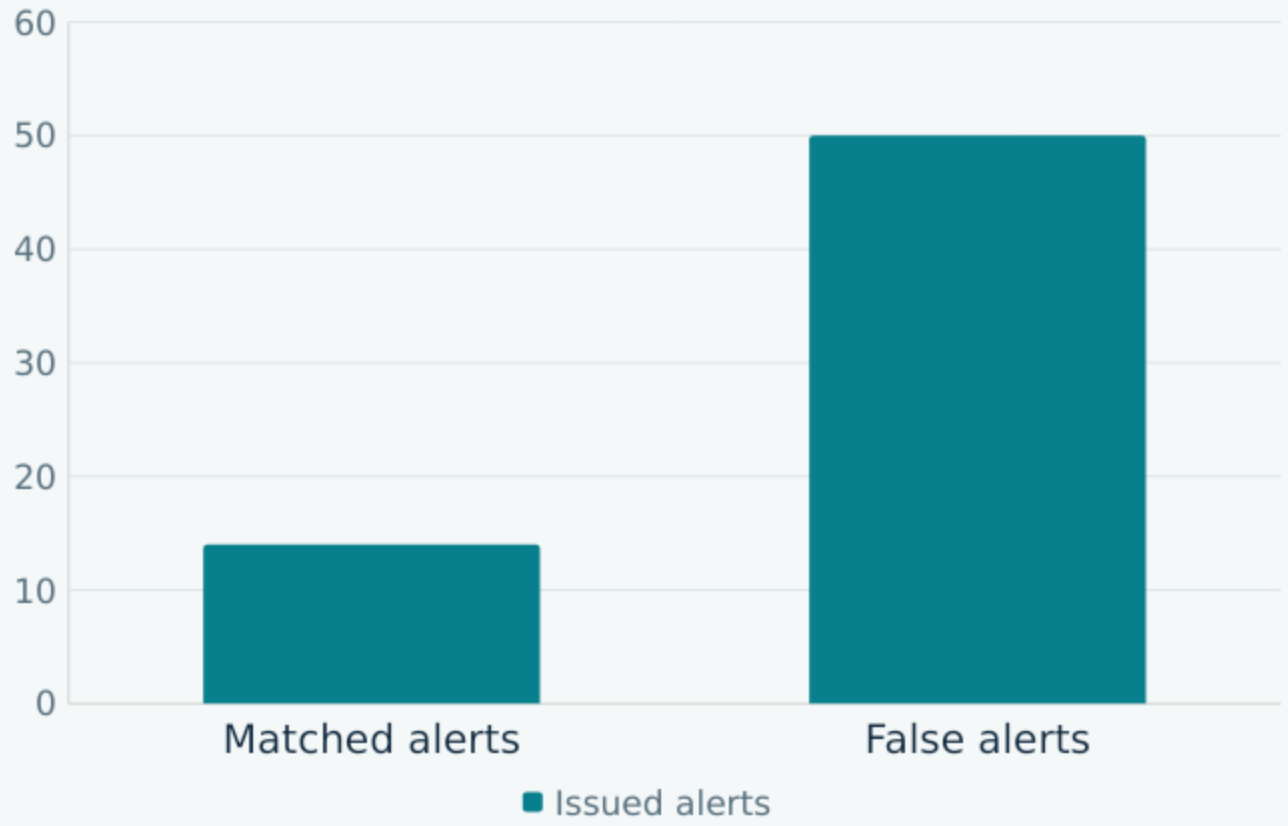
Review

Forecast + uncertainty + data age.

Inspect

Reveal future only for evaluation.

The alert experiment needs improvement



14 / 66

observable onsets detected

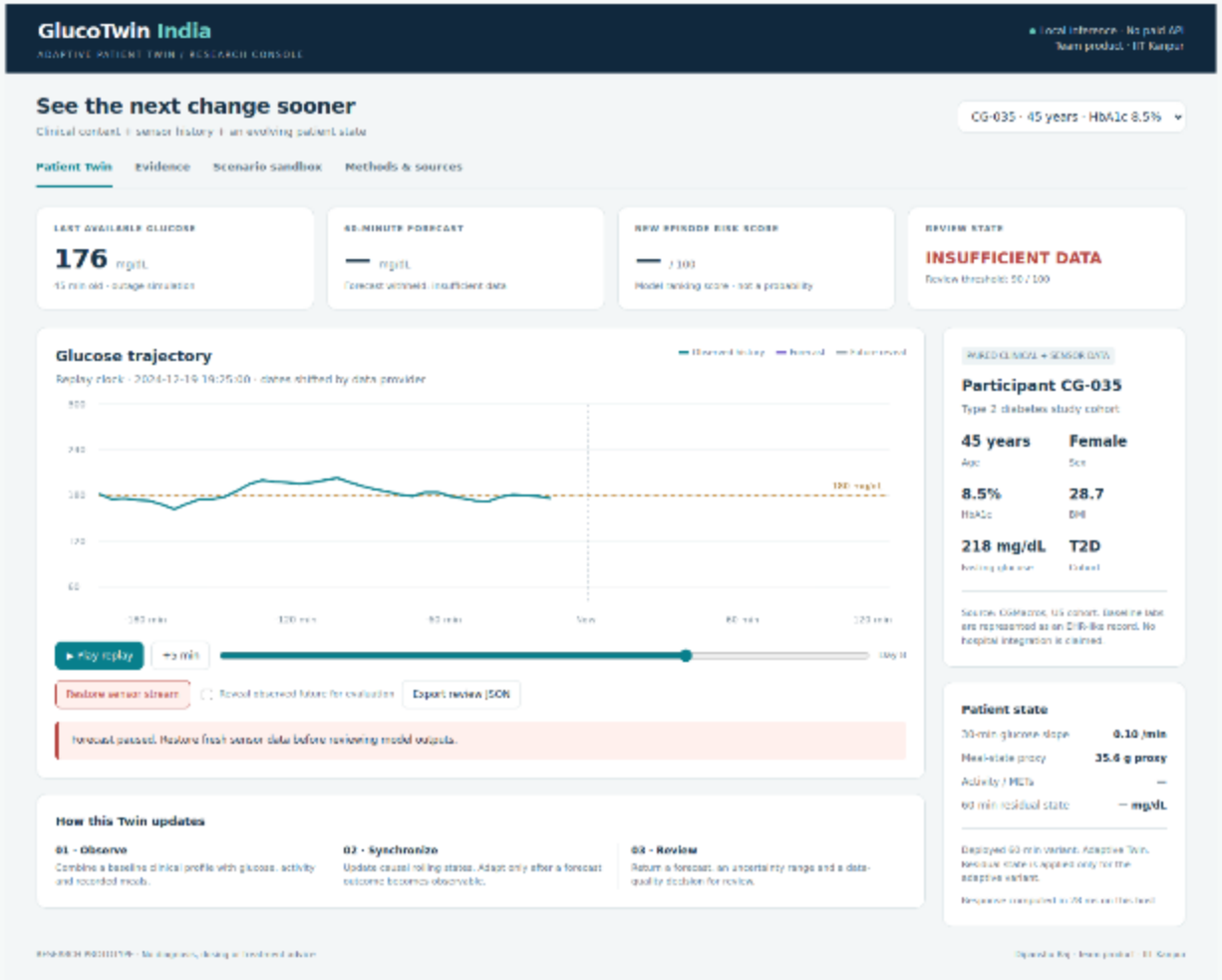
15 min

median matched warning

21.9% precision · 21.2% recall · 1.87 false alerts / eligible patient-day

Exploratory review flag. These results do not support clinical reliance.

Failure handling is part of the prototype



Withhold

No forecast when inputs are inadequate.

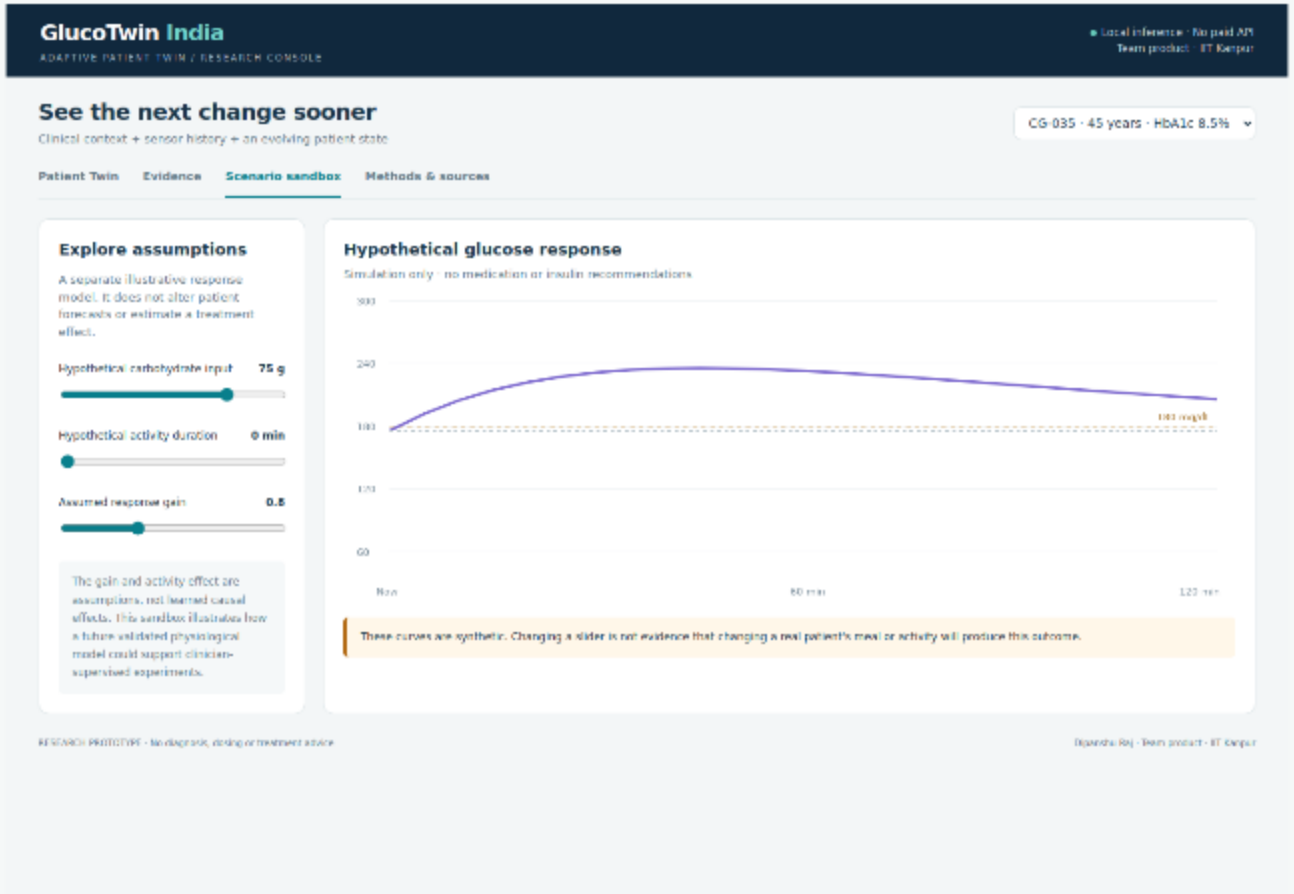
Show age

Historical values stay labeled as stale.

Recover

Restore outputs when quality conditions pass.

A clearly labeled scenario sandbox



Assumptions, not treatment effects

Carbohydrate input
Activity duration
Response gain

Synthetic curves are excluded from the real-data evaluation.

India deployment is a validation pathway

01 Native data

Obtain appropriately governed CGM exports and matched Indian clinical records.

02 External validation

Preregister endpoints; evaluate subgroup performance, sensors and missingness.

03 Better event risk

Improve discrimination, calibrate scores and agree on an acceptable alert burden.

04 Prospective silent study

Evaluate before changing care; assess safety and regulatory requirements.

A compact, reproducible implementation

No GPU. No paid API.

Component	Implementation
Forecasting	Python + scikit-learn histogram gradient boosting
State update	Causal rolling features + delayed residual memory
Interface	Local Python server + HTML/CSS/JavaScript
Reproduction	Source CSVs, trained model, exact splits and tests

12 core tests passed

Desktop and mobile interactions checked

Training: 26.8 seconds on the build host; timing is hardware-dependent.

What this submission establishes

A working twin.
An auditable experiment.

Paired data · Evolving state · Measured forecasts
Visible uncertainty · Explicit failure handling

Next: larger Indian validation and stronger event modeling

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